

VERHNYAYA TOM, Russian Federation

WMO: 314590

Lat: 51.350N

Lon: 130.433E

Elev: 856

StdP: 14.25

Time Zone: 9.00 (E09)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.1	-40.5	-48.7	0.3	-43.6	-45.2	0.4	-40.2	14.2	7.6	12.4	5.1	0.1	N/A	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.8	84.3	69.8	81.4	68.4	78.5	66.9	73.1	80.4	71.0	78.4	69.0	75.9	5.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.9	117.6	77.6	68.4	107.9	75.3	66.5	100.7	73.2	37.5	80.5	35.6	79.1	33.8	75.7	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	14.3	11.8	-48.2	88.5	4.7	2.4	-51.6	90.3	-54.4	91.6	-57.0	93.0	-60.5	94.7		
(5)			-47.4	76.8	3.8	2.1	-50.2	78.3	-52.4	79.5	-54.6	80.7	-57.3	82.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	-19.7	-9.3	9.4	33.6	48.4	60.3	65.6	61.3	48.7	30.8	4.1	-15.0		
	DBStd	31.30	8.79	10.03	11.15	8.34	7.68	5.54	4.94	5.24	7.56	9.57	12.01	9.88		
	HDD50	9801	2160	1659	1258	495	120	2	0	0	117	597	1377	2016		
	HDD65	14097	2625	2079	1723	942	515	160	54	138	491	1062	1827	2481		
	CDD50	1294	0	0	0	2	72	311	482	350	76	1	0	0		
	CDD65	116	0	0	0	0	2	19	71	23	1	0	0	0		
	CDH74	1830	0	0	0	0	97	487	958	272	16	0	0	0		
	CDH80	382	0	0	0	0	10	105	230	36	1	0	0	0		
(14) Wind	WSAvg	2.3	0.9	1.2	2.4	3.7	3.3	2.8	2.4	2.5	3.0	2.7	1.8	0.6		
(15) Precipitation	PrecAvg	30.1	0.4	0.4	0.9	1.7	3.1	4.1	5.8	5.4	3.6	2.4	1.4	0.8		
	PrecMax	40.9	0.9	1.1	1.9	3.7	6.4	7.6	9.8	8.3	6.4	5.2	2.9	1.8		
	PrecMin	22.8	0.1	0.0	0.1	0.5	1.2	1.4	1.1	2.1	1.3	0.6	0.4	0.3		
	PrecStd	4.3	0.2	0.3	0.5	0.8	1.3	1.4	1.9	1.6	1.3	0.9	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	26.9	49.9	67.5	81.5	87.0	88.9	84.1	76.3	62.5	41.0	20.8		
		MCWB	11.7	22.9	37.7	50.2	59.5	69.1	72.2	72.9	62.7	50.9	34.5	18.8		
	2%	DB	8.7	20.8	41.1	60.6	77.2	83.5	85.4	79.9	71.4	57.2	32.0	14.3		
		MCWB	7.4	17.2	32.7	46.0	57.0	67.7	70.8	69.4	60.3	45.8	28.5	12.9		
	5%	DB	3.7	15.7	35.7	55.8	72.4	80.1	82.4	76.9	67.7	52.3	28.0	8.4		
		MCWB	2.6	12.9	29.0	42.8	55.7	65.7	69.9	68.2	59.4	43.0	24.9	7.1		
	10%	DB	-1.1	11.2	31.2	51.2	67.6	76.4	79.7	74.3	64.0	47.7	24.3	3.0		
		MCWB	-2.0	9.0	25.9	40.5	53.2	63.8	69.5	67.0	57.3	40.6	21.7	2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	23.1	39.4	50.8	61.6	74.0	76.7	75.4	66.0	52.3	33.9	18.7		
		MCDB	13.5	27.2	49.4	65.9	74.5	83.2	83.4	80.7	71.1	59.9	40.5	20.3		
	2%	WB	7.5	17.3	33.5	46.6	59.4	70.0	74.0	71.9	63.2	47.6	28.8	13.0		
		MCDB	8.7	20.6	39.6	57.3	73.0	79.8	81.2	77.4	68.6	54.4	31.6	14.2		
	5%	WB	2.7	13.0	29.5	43.8	57.0	67.6	72.1	69.4	60.7	44.4	25.0	7.1		
		MCDB	3.8	15.5	35.4	53.9	70.4	76.9	79.9	74.5	65.5	51.1	27.9	8.3		
	10%	WB	-2.1	9.0	26.0	41.1	54.3	65.3	70.3	67.6	57.8	40.8	21.8	2.1		
		MCDB	-1.2	11.2	31.1	50.1	65.7	73.9	77.7	73.1	62.6	47.3	24.3	2.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	29.5	35.1	34.8	25.7	28.3	27.1	24.8	23.2	24.6	25.0	26.0	26.2		
		MCDBR	30.3	36.5	35.4	33.3	37.6	34.4	30.5	28.4	30.0	33.5	25.6	28.4		
	5% WB	MCWBR	28.9	33.2	27.5	20.8	21.2	20.1	17.7	19.0	20.1	23.1	22.2	27.0		
		MCWBR	30.3	35.6	33.5	30.3	33.0	28.7	26.2	24.6	24.5	29.5	24.6	28.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.262	0.297	0.366	0.537	0.587	0.493	0.473	0.419	0.395	0.417	0.328	0.265		
		taud	2.134	2.055	1.913	1.692	1.690	2.023	2.134	2.283	2.251	1.999	2.066	2.090		
	Ebn at Noon	Ebn at Noon	235	256	255	221	218	245	248	257	248	204	198	209		
		Edh at Noon	23	34	48	69	72	52	46	37	35	36	26	21		
(44) All-Sky Solar Radiation	RadAvg	496	874	1385	1603	1624	1787	1659	1460	1118	728	490	385			
	RadStd	15	21	66	150	145	223	132	103	106	51	28	18			

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page